Table 1: E. coli + P. putida datasets (pre-build, 2026-10-15) built as LMDBs in the TorchCell dev tree, with a description of each. *Description* is the summary line of the dataset class's docstring; the other columns are as in the non-expanded table.

Dataset	Host	Description	Genotypes	Env	Instances	Phenotype	Shape	Graph role	Signal (gzip, bits)
Environmental / chemogenomic response									
CrisprPineneToleranceNiu2019	E. coli	Niu 2019 CRISPRa/CRISPRi growth ratios under 0.5% pinene in BW25113(PT5-dxs).	54	1	54	environment_response (log2_ratio)	scalar	global	4.0×10^{4}
CrispriChemgenChoe2025	E. coli	Choe 2025 per-guide CRISPRi abundance response to twelve antibiotics.	39,291	12	470,502	environment_response (log2_ratio)	scalar	global	2.4×10^{8}
CrispriGeneFitnessWang2018	E. coli	Wang 2018 gene-level CRISPRi calls (Supplementary Data 5) across five screens.	13,641	5	20,158	environment_response (log2_ratio)	scalar	global	1.5×10^{7}
CrispriGuideFfaEnrichmentFang2025	E. coli	Fang 2025 per-guide CRISPRi-FACS enrichment across the paper's two rounds.	15,711	1	15,711	environment_response (log2_ratio)	scalar	global	7.3×10^{6}
CrispriGuideFitnessWang2018	E. coli	Wang 2018 per-guide CRISPRi fitness across the paper's five pooled screens.	53,203	5	240,481	environment_response (log2_ratio)	scalar	global	1.6×10^{8}
CrispriKnockdownCui2018	E. coli	Cui 2018 per-guide dCas9 knockdown log2 fold changes in two MG1655 strains.	71,696	1	143,392	environment_response (log2_ratio)	scalar	global	7.1×10^{7}
CrispriScreenRousset2018	E. coli	Rousset 2018 per-sgRNA log2FC: growth, three phage challenges, transduction.	40,592	4	92,252	environment_response (log2_ratio)	scalar	global	4.1×10^{7}
DoublingTimeCaglar2017	E. coli	Caglar 2017 Table S5: 55 absolute doubling times, one per replicate curve.	1	16	55	environment_response (growth_rate)	scalar	global	3.2×10^{4}
EnvChemgenGirgis2009	E. coli	Combined z-scores of MG1655 transposon mutants in 17 antibiotics.	3,976	17	66,287	environment_response (z_score)	scalar	global	3.6×10^{7}
EnvChemgenNichols2011	E. coli	Nichols 2011 Table S2: 3,979 E. coli strains' re-scaled scores in 324 screens.	3,967	324	1,286,962	environment_response (z_score)	scalar	global	4.0×10^{8}
EnvChemgenShiver2016	E. coli	Shiver 2016 KEIO deletion fitness-scores across 57 chemical-genomic conditions.	3,846	51	210,998	environment_response (z_score)	scalar	global	9.9×10^{7}
EnvChemgenWang2015	E. coli	Isoprenol tolerance of 46 Keio transporter deletions (Wang 2015, Table S3).	46	1	46	environment_response (log2_ratio)	scalar	global	2.9×10^{4}
EnvChemgenWang2024	E. coli	TRANSIT log2FC of MG1655 Tn5 mutants at three rifampicin doses x two times.	4,419	6	26,514	environment_response (log2_ratio)	scalar	global	9.3×10^{6}
FitnessBrowserEcoli	E. coli	Fitness Browser Keio experiments not loaded from another release, by publication.	3,789	3	22,734	environment_response (log2_ratio)	scalar	global	1.1×10^{7}
GrowthRateLamoureux2023	E. coli	PRECISE-1K's released per-sample growth rate: 354 absolute rates in 1/hr.	21	54	354	environment_response (growth_rate)	scalar	global	1.9×10^{5}
GrowthRateS23Schmidt2016	E. coli	Schmidt 2016 Table S23: 24 absolute per-condition growth rates, three strains.	1	22	26	environment_response (growth_rate)	scalar	global	2.0×10^{4}
GrowthYieldBaba2006	E. coli	Baba 2006 Table 3: 11,735 released OD600 readings of obtained Keio mutants.	3,912	3	11,735	environment_response (optical_density)	scalar	global	3.3×10^{6}
MismatchCrispriFitnessHawkins2020	E. coli	Hawkins 2020 per-sgRNA relative fitness of E. coli essential-gene knockdowns.	24,186	1	24,186	environment_response (relative_growth_rate)	scalar	global	1.5×10^{7}
PhageRbTnseqMutalik2020	E. coli	Mutalik 2020 per-(gene, experiment) phage-challenge RB-TnSeq gene fitness.	3,716	71	287,815	environment_response (log2_ratio)	scalar	global	2.1×10^{8}
RbTnseqPrice2018Ecoli	E. coli	Price 2018 compendium, E. coli BW25113 KEIO_ML9: gene fitness per sample.	3,789	104	598,662	environment_response (log2_ratio)	scalar	global	4.0×10^{8}
RbTnseqTeteneva2024	E. coli	Teteneva 2024, E. coli K-12 W3110 RpoS+ in lake water: Table S4 gene fitness.	3,691	18	66,162	environment_response (log2_ratio)	scalar	global	2.3×10^{7}
RbTnseqWetmore2015	E. coli	Wetmore 2015, E. coli BW25113 KEIO_ML9: gene fitness per successful sample.	3,646	85	353,662	environment_response (log2_ratio)	scalar	global	2.4×10^{8}
EnvMetalTnseqRoyet2025	P. putida	KT2440 mariner Tn-seq log2FC of every gene in cobalt, copper, zinc and cadmium.	5,491	4	21,583	environment_response (log2_ratio)	scalar	global	6.7×10^{6}
FitnessBrowserPutida	P. putida	Fitness Browser KT2440 experiments of the attributed studies, by publication.	9,556	146	1,118,052	environment_response (log2_ratio)	scalar	global	5.3×10^{8}
IsoprenolSelectionMenasalvas2025	P. putida	Menasalvas 2025 enriched CRISPRi knockdowns from the growth-coupled selection.	58	1	58	environment_response (categorical)	scalar	global	7.5×10^{4}
IsoprenolToleranceLim2025	P. putida	Isoprenol tolerance of IPL300, IPL400 and KT2440 dPP_3024 (Lim 2025).	3	2	3	environment_response (log2_ratio)	scalar	global	1.8×10^{4}
LactamGrowthRateThompson2019	P. putida	Thompson 2019 Table S1: maximal growth rates on three carbon sources.	3	3	9	environment_response (growth_rate)	scalar	global	1.2×10^{4}
RbTnseqBorchert2023	P. putida	Borchert 2023 KT2440 RB-TnSeq fitness for the 271 loci the compendium dropped.	271	32	10,824	environment_response (log2_ratio)	scalar	global	4.5×10^{6}
RbTnseqBorchert2024	P. putida	KT2440 RB-TnSeq gene fitness, 4,732 genes x the kept samples of 332.	9,464	155	1,466,920	environment_response (log2_ratio)	scalar	global	4.2×10^{8}
Expression (RNA-seq)									
MrnaFractionBalakrishnan2022	E. coli	NCM3722-family mRNA number fractions, one record per released sample column.	3	14	28	mrna_number_fraction (rnaseq_mrna_number_fraction)	vector (4342)	node	7.2×10^{6}
RnaseqCaglar2017	E. coli	Caglar 2017 RNA-seq: one record per REL606 library (Table S2 sample).	1	57	152	expression_tpm (rnaseq_tpm)	vector (4196)	node	6.7×10^{7}
RnaseqGuzman2018	E. coli	Guzmán 2018's 14 released RNA-seq libraries, from its own GEO release.	7	2	14	expression_tpm (rnaseq_tpm)	vector (4329)	node	4.5×10^{6}
RnaseqLamoureux2023	E. coli	PRECISE-1K: one record per MG1655 RNA-seq library (wild type or deletions).	74	119	1,035	expression_tpm (rnaseq_tpm)	vector (4257)	node	5.8×10^{8}
RnaseqPublicK12Lamoureux2023	E. coli	Public K-12: one record per reprocessed public MG1655 RNA-seq library.	72	198	1,675	expression_tpm (rnaseq_tpm_from_released_counts)	vector (4355)	node	9.5×10^{8}
PutidaPrecise321Lim2022	P. putida	putidaPRECISE321: per-sample absolute KT2440 transcriptomes (TPM + counts).	14	118	321	expression_tpm (rnaseq_tpm)	vector (5564)	node	2.4×10^{8}
Fitness									
CarbonSourceTong2020	E. coli	Tong 2020 Keio and sRNA/small protein deletion growth on 30 carbon sources.	3,714	30	111,420	fitness	scalar	global	4.8×10^{7}
CrispriCrossRachwalski2024	E. coli	Rachwalski 2024 normalized colony growth of CRISPRi knockdown by deletion crosses.	14,209	15	48,900	fitness	scalar	global	1.9×10^{7}
GrowthAucRapp2026	E. coli	Growth of Rapp 2026's CRISPRi library, as the paper's AUC over the controls.	1,515	1	1,515	fitness	scalar	global	7.7×10^{5}
GrowthRateCampos2018	E. coli	Campos 2018 Keio maximal-growth-rate fitness in M9 casamino-acid glucose.	3,664	1	3,664	fitness	scalar	global	1.1×10^{6}
GrowthRateChoe2019	E. coli	Choe 2019 Fig. 2c: the two MS56 designed deletions, as parent-relative fitness.	6	1	8	fitness	scalar	global	1.1×10^{4}
GrowthRateSchmidt2016	E. coli	Schmidt 2016 Table S24: rim-deletion growth rates as WT-relative fitness.	3	2	6	fitness	scalar	global	7.8×10^{3}
GrowthWang2015	E. coli	Plain-medium growth of 46 Keio transporter deletions (Wang 2015, Table S3).	46	1	46	fitness	scalar	global	2.1×10^{4}
TranscriptionFactorKnockoutChoe2019	E. coli	Choe 2019 Supplementary Fig. 6: the two Keio deletions that carry a number.	2	1	2	fitness	scalar	global	5.3×10^{3}
FitnessContrastBorchert2023	P. putida	Borchert 2023's 64,853 released two-arm fitness contrasts.	5,031	12	64,853	t_statistic (rbtnseq_gene_fitness_two_sample_t_test)	scalar	global	2.2×10^{7}
Flux									
FluxIshii2007	E. coli	Fitted 13C net-flux maps of 28 chemostat cultures.	25	5	28	net_flux (c13_mfa_net_flux_percent_of_specific_glucose_uptake)	vector (42)	bipartite node	6.0×10^{4}
FluxLi2021	E. coli	Li 2021's three released 13C-MFA maps, against BW-PBF's unreleased map.	3	1	3	net_flux (c13_mfa_net_flux_mmol_per_gdw_per_h)	vector (99)	bipartite node	5.7×10^{4}
Gene interaction									
GeneInteractionBabu2014	E. coli	eSGA interaction scores of E. coli K-12 double deletion mutants (Table S2).	42,703	1	42,705	gene_interaction	scalar	edge	1.2×10^{7}
GeneInteractionButland2008	E. coli	Unfiltered eSGA interaction scores of E. coli K-12 double deletion mutants.	314,769	1	314,769	gene_interaction	scalar	edge	8.2×10^{7}
Metabolite									
ExometaboliteBrunk2016	E. coli	The six HPLC columns the workbook labels g/L: glucose and five acids.	9	28	126	metabolite_level (hplc_extracellular_concentration_g_per_l)	vector (6)	bipartite node	1.8×10^{5}
MetaboliteIntensityRapp2026	E. coli	Absolute FI-MS intensity of Rapp 2026's accumulating annotated features.	407	1	407	metabolite_level (fi_ms_annotated_feature_absolute_intensity_mean_of_2_plates)	vector (2)	bipartite node	3.9×10^{5}
MetabolomeBrunk2016	E. coli	The 51 LC-MS columns the workbook labels (uM), one record per sample.	9	26	117	metabolite_level (lc_ms_concentration_um)	vector (51)	bipartite node	3.5×10^{5}
MetabolomeFuhrer2017	E. coli	Genome-wide FIA-TOF-MS ion metabolome of the E. coli Keio deletion collection.	3,806	1	3,806	metabolite_level (fia_tof_ms_ion_modified_z_score)	vector (7534)	bipartite node	1.9×10^{9}
MetabolomeIshii2007	E. coli	CE-TOFMS intracellular metabolite concentrations of 28 chemostat cultures.	25	5	28	metabolite_level (ce_tofms_intracellular_concentration_mm)	vector (130)	bipartite node	4.9×10^{5}
MetabolomeRapp2026	E. coli	FI-MS metabolome of a CRISPRi library covering every iML1515 gene of E. coli.	1,497	1	1,497	metabolite_level (fi_ms_iml1515_feature_fold_change_vs_batch_median_mean_of_2_plates)	vector (1321)	bipartite node	5.6×10^{8}
MetabolomeSchastnaya2021	E. coli	FIA-TOF-MS metabolome of the KEIO deletion arm of an E. coli phosphosite screen.	103	5	200	metabolite_level (fia_tof_ms_ion_log2_fold_change_vs_wild_type)	vector (284)	bipartite node	3.6×10^{6}
TargetedMetabolomeRapp2026	E. coli	Targeted LC-MS/MS of Rapp 2026's accumulating strain-metabolite pairs.	407	1	407	metabolite_level (lc_msms_eic_peak_height_fold_change_vs_library_median)	vector (2)	bipartite node	3.7×10^{5}
MetaboliteGrowthPhaseMenasalvas2025	P. putida	data S1-1's growth-phase (GP) absolute metabolite concentrations.	3	1	3	metabolite_level (lc_ms_intracellular_concentration_uM)	vector (46)	bipartite node	7.4×10^{4}
MetaboliteProductionPhaseMenasalvas2025	P. putida	data S1-1's production-phase (PP) absolute metabolite concentrations.	3	1	3	metabolite_level (lc_ms_intracellular_concentration_uM)	vector (46)	bipartite node	7.3×10^{4}
Morphology									
MorphologyCampos2018	E. coli	Campos 2018 Keio cell-morphology profiles: 26 features per imaged strain.	3,664	1	3,664	morphology	vector (15)	global	8.5×10^{6}
Phosphoproteome									
PhosphositeIntensityPotel2018	E. coli	Potel 2018 phosphosite intensities of W3110 in 12 released samples.	1	2	12	phosphosite_intensity (fe3_imac_lc_msms_maxquant_summed_xic)	vector (989)	node	1.3×10^{7}
Product titer									
BiofuelTiterBrunk2016	E. coli	Brunk 2016 isopentenol, limonene and bisabolene titers over the time course.	8	18	72	titer	scalar	global	1.1×10^{5}
IsopentenolTiterFoo2014	E. coli	Foo 2014 isopentenol titers: the eight tolerance strains and the RFP control.	9	1	9	titer	scalar	global	1.5×10^{4}
PineneFoldChangeNiu2019	E. coli	Niu 2019 CRISPRa/CRISPRi pinene production ratios in BW25113(PT5-dxs).	40	1	40	titer	scalar	global	2.2×10^{4}
IsoprenolTiterBanerjee2024	P. putida	Banerjee 2024 isoprenol titers: the 20 the article body states numerically.	14	9	20	titer	scalar	global	3.4×10^{4}
IsoprenolTiterCarruthers2025	P. putida	Carruthers 2025 per-strain isoprenol titer across six DBTL cycles.	432	2	502	titer	scalar	global	4.2×10^{5}
IsoprenolTiterDeSiqueira2025	P. putida	de Siqueira 2025 maximum isoprenol titer of PT carrying pIY670.	5	2	10	titer	scalar	global	2.1×10^{5}
IsoprenolTiterYunus2026	P. putida	Yunus 2026 per-strain isoprenol titer, from the hand-deposited Benchling table.	125	1	125	titer	scalar	global	6.6×10^{4}
IsoprenylAcetateTiterKang2026	P. putida	Kang 2026 isoprenyl acetate titers: Table 1, Table S4 and the Table S9 fed-batch.	11	13	26	titer	scalar	global	6.7×10^{4}
ValerolactamTiterThompson2019	P. putida	Thompson 2019 valerolactam titers: the four-strain ladder at 24 and 48 h.	4	2	8	titer	scalar	global	1.7×10^{4}
Promoter activity									
PromoterReporterMohiuddin2022	E. coli	Mohiuddin 2022 hourly promoter-GFP readings under three antibiotics.	1,809	27	69,480	promoter_activity	scalar	node	2.8×10^{7}
Protein abundance									
ProteinFoldChangeCaglar2017	E. coli	Caglar 2017 differential proteomics: one record per Table S8 protein contrast.	1	8	8	protein_fold_change (deseq2_wald_log2_fold_change_design_batch_plus_condition)	vector (4180)	node	1.1×10^{7}
ProteomeBrunk2016	E. coli	Brunk 2016 SRM peak areas: nine strains at nine hours, all 68 host proteins.	9	18	81	protein_abundance (srm_protein_peak_area)	vector (68)	node	3.6×10^{5}
ProteomeCaglar2017	E. coli	Caglar 2017 proteome: one record per REL606 LC-MS/MS sample (Table S3).	1	40	105	protein_abundance (lcmsms_spectral_count_deseq2_size_factor_normalized)	vector (4196)	node	2.5×10^{7}
ProteomeIshii2007	E. coli	LC-MS/MS absolute protein abundances of 28 chemostat cultures.	25	5	28	protein_abundance (lc_ms_ms_absolute_mg_protein_per_g_dry_cell_weight)	vector (57)	node	2.9×10^{5}
ProteomeMori2021	E. coli	Absolute MG1655 proteome mass fractions, one record per loaded sample.	1	1	7	protein_abundance (absolute_protein_mass_fraction_xtop_dia_swath_riboprofiling_scaled)	vector (1897)	node	1.1×10^{6}
ProteomeSchmidt2016	E. coli	Absolute BW25113 proteome, one record per loaded growth condition.	1	21	21	protein_abundance (absolute_protein_copies_per_cell_label_free_sid_anchored)	vector (2359)	node	1.1×10^{7}
ProteomeSrmSet1Schmidt2016	E. coli	Table S2: the OGE-fractionated arm's SRM absolute abundances, 11 records.	1	18	18	protein_abundance (absolute_protein_copies_per_cell_srm_sid_dataset1_technical_sd)	vector (41)	node	1.5×10^{5}
ProteomeSrmSet2Schmidt2016	E. coli	Table S3: the unfractionated arm's SRM absolute abundances, 14 records.	1	21	21	protein_abundance (absolute_protein_copies_per_cell_srm_sid_dataset2_biological_sd)	vector (31)	node	1.3×10^{5}
CampaignProteomeCarruthers2025	P. putida	Carruthers 2025 campaign proteome: one record per CRISPRi strain-cycle.	401	1	465	protein_abundance (dia_nn_top3_percent_of_proteome_mean)	vector (1616)	node	1.3×10^{8}
CrispriArrayYunus2026	P. putida	Yunus 2026 multiplexed-array relative expression panels (Tables S8-S12).	25	1	25	protein_fold_change (dia_nn_top3_relative_to_control_strain)	scalar	node	2.5×10^{4}
CrispriDifferentialProteomeYunus2026	P. putida	Yunus 2026 global fold change against the control for the PP_4188 strain.	1	1	1	protein_fold_change (dia_nn_top3_fold_change_relative_to_control_strain_equal_variance_t_test)	vector (305)	node	6.1×10^{4}
CrispriKnockdownYunus2026	P. putida	Yunus 2026 per-strain relative expression of its own CRISPRi target (Table S3).	102	1	102	protein_fold_change (dia_nn_top3_relative_to_control_strain)	scalar	node	4.9×10^{4}
CrispriPanelProteomeYunus2026	P. putida	Yunus 2026 per-strain Top3 protein panel, from the hand-deposited table.	125	1	125	protein_abundance (dia_nn_top3_signal_benchling_displayed)	vector (207)	node	1.5×10^{6}
ProteomeBanerjee2025	P. putida	Banerjee 2025 DIA proteome: six released samples of the p-CA cutset strains.	4	3	6	protein_abundance (diann_lfq_log2_intensity_replicate_mean)	vector (2248)	node	2.8×10^{6}
ProteomeCarruthers2025	P. putida	Carruthers 2025 Top3 proteome of the PP_0815 off-target-study strains.	17	2	21	protein_abundance (dia_nn_top3_peptide_signal_mean)	vector (1424)	node	5.6×10^{6}
ProteomeDeSiqueira2025	P. putida	de Siqueira 2025 Top3 proteome of the writable KT2440 and PT strains.	6	3	17	protein_abundance (dia_nn_top3_peptide_signal_replicate_mean)	vector (1531)	node	5.5×10^{6}
ProteomeFoldChangeCarruthers2025	P. putida	Carruthers 2025 per-protein log2 fold changes of the CRISPRi and KO contrasts.	16	1	16	protein_fold_change (dia_log2_fold_change_paired_two_tailed_t_test)	vector (350)	node	1.5×10^{6}
ProteomeFoldChangeLim2025	P. putida	Lim 2025's four released evolved-isolate-over-IPL400 proteome contrasts.	2	2	4	protein_fold_change (dia_nn_top3_log2_fold_change_welch_two_sample_t_test)	vector (2361)	node	3.7×10^{6}
ProteomeLim2025	P. putida	IPL400 and its two evolved isolates under 4 g/L isoprenol (Lim 2025).	3	1	3	protein_abundance (dia_nn_top3_log2_mean)	vector (2361)	node	1.5×10^{6}
ProteomeLog10PercentDeSiqueira2025	P. putida	de Siqueira 2025 mean log10 percent-of-total abundance, the third released scale.	6	3	17	protein_abundance (dia_nn_top3_log10_percent_of_total_abundance_replicate_mean)	vector (1531)	node	6.0×10^{6}
ProteomeMenasalvas2025	P. putida	Menasalvas 2025 growth-phase and production-phase proteome of the three strains.	3	2	6	protein_abundance (dia_protein_counts_sum_mean)	vector (2090)	node	2.6×10^{6}
ProteomePercentDeSiqueira2025	P. putida	de Siqueira 2025 percent-of-total relative abundance, the paper's own scale.	6	3	17	protein_abundance (dia_nn_top3_percent_of_total_abundance_replicate_mean)	vector (1531)	node	6.0×10^{6}
Protein synthesis / turnover									
ProteinSynthesisRateLi2014	E. coli	Li 2014 absolute protein synthesis rates of MG1655 in three MOPS media.	1	3	3	synthesis_rate (ribosome_profiling_footprint_density_absolute_synthesis_rate)	vector (3041)	node	3.7×10^{5}
ProteinTurnoverGupta2024	E. coli	Gupta 2024 per-protein total turnover in 13 NCM3722 growth conditions.	6	8	13	degradation_rate (n15_ammonium_tmtproc_total_turnover_rate_per_hour)	vector (2553)	node	1.1×10^{7}
Viability									
GeneEssentialityBaba2006	E. coli	Keio construction outcomes of 4,288 targeted E. coli BW25113 ORFs.	4,288	1	4,288	is_essential	scalar	node	1.8×10^{6}
GeneEssentialityGerdes2003	E. coli	Genetic-footprinting essentiality calls of E. coli MG1655 genes (Table S1).	3,739	1	3,739	is_essential	scalar	node	1.9×10^{6}
GeneEssentialityGoodall2018	E. coli	TraDIS essentiality calls of E. coli BW25113 genes, as plated and after LB.	8,309	2	8,309	is_essential	scalar	node	5.1×10^{6}
GeneEssentialityPrice2018Ecoli	E. coli	Price 2018 Supplementary Table 1: E. coli BW25113 likely-essential genes.	324	1	324	is_essential	scalar	node	2.6×10^{5}
GradedGeneEssentialityGoodall2018	E. coli	Goodall 2018's 317 "unclear" TraDIS calls, stored as released (#909).	317	2	317	essentiality_call	scalar	node	2.2×10^{5}
GradedGeneEssentialityRoyet2025	P. putida	Royet 2025 Table S4: every KT2440 gene's TRANSIT HMM state on LB agar and after outgrowth in LB, stored as released (#909).	5,729	2	11,458	essentiality_call	scalar	node	4.8×10^{6}
Total (100 datasets, 100 probed)					**7,257,468**				7.7×10^{9}